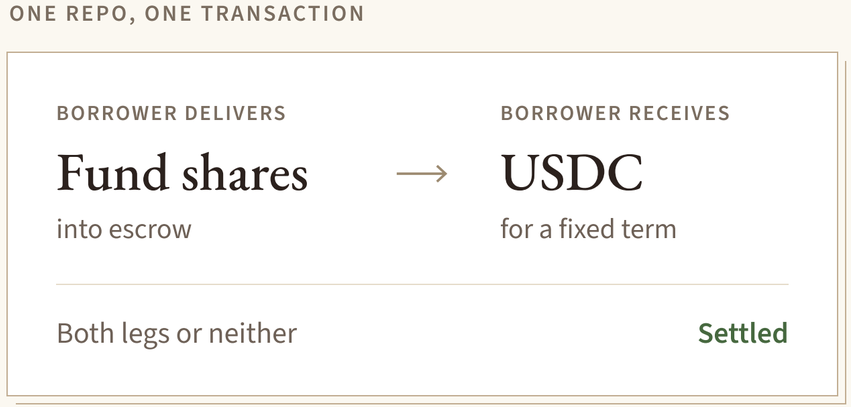
Caravel

Repo for RWAs on Stellar

Fixed-rate USDC loans against tokenized Treasury and money-market funds, priced off daily NAV, settled atomically.

Per aspera ad Stellar



HackMeridian 2026 · Lisbon · Built on Stellar testnet

Billions in fund shares sit on Stellar. Almost none of it works.

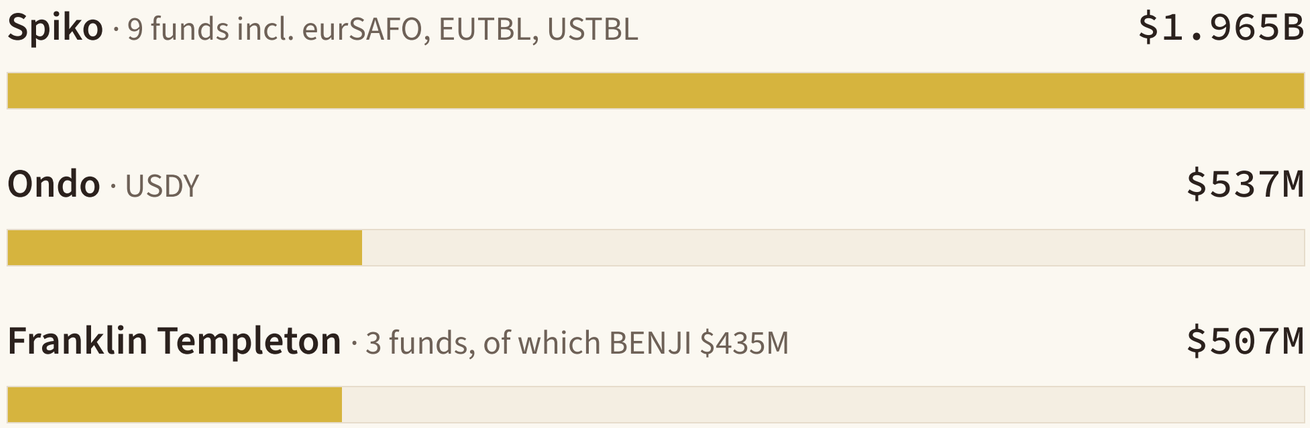
$3.03B

RWAs on Stellar (Active AUM)

$133K

of that is used in DeFi (DeFi Active TVL)

LARGEST ISSUERS ON STELLAR



0 smart contracts hold BENJI today.
No on-chain lender can take it as collateral.

Sources, as of 25 Oct 2026: DefiLlama RWA, Stellar chain page, default filters (Active AUM, DeFi Active TVL, issuers); DefiLlama BENJI asset page (Stellar share); Stellar Horizon, BENJI asset holders.

A 24/7 liquidation bot can’t lend against a fund share

PRICE

Priced once a day

Funds publish NAV on business days. Bots liquidate every block, even at **3 a.m.** on a Sunday.

EXIT

Nowhere to sell

eurSAFO and EUTBL: **0 trades** on Stellar. EUTBL “stopped trading on all exchanges listed” (CoinGecko).

ACCESS

Approved holders

The tokens are permissioned. A permissionless bot **can’t even receive** the seized shares.

The asset is safe. Turning it back into cash takes business days, through the issuer.

Sources, as of 25 Oct 2026: stellar.expert trade history for eurSAFO and EUTBL; DefiLlama eurSAFO token page ($0 DEX volume); CoinGecko (Spiko EU T-Bills MMF).

Fixed-term loans are the only safe way to lend against collateral that takes days to redeem.

At open

NAV sets the collateral. Both legs settle together.

Once a day

Margin check on the new NAV. A breach opens a top-up window.

At maturity

Repay, or default to a liquidator who redeems with the issuer.

A price is needed only three times. Money markets already lend against such assets this way: it is called repo.

Source: Caravel collateral research, §1 (exit, not the asset, is the weak link) and §8 (a term that matches redemption time).

One repo, from offer to exit

1
Offer
Lender escrows USDC at a fixed rate and term.

→

2
Atomic DvP
Shares in, USDC out, one transaction. Both legs or neither.

→

3
Daily margin
Keeper checks the NAV. On a breach, the borrower tops up.

→

4
Repay
Borrower pays the fixed debt and gets the shares back.

missed deadline ↓

4b
Default
Anyone can call it once maturity or the margin window passes.

→

5
Liquidate
A whitelisted liquidator buys at NAV − 2% and redeems with the issuer.

Lenders never hold fund shares.
USDC in, USDC out. Only the contract, borrowers and liquidators need the issuer’s approval.

Source: Caravel README, “How it works”; DECISIONS #7, #14, #15 (2% liquidation discount, debt fixed at take).

Live on Stellar testnet

1 TAKE: THE DVP RECEIPT

BORROWER DELIVERS
103.20 tBENJI
into escrow

⇄

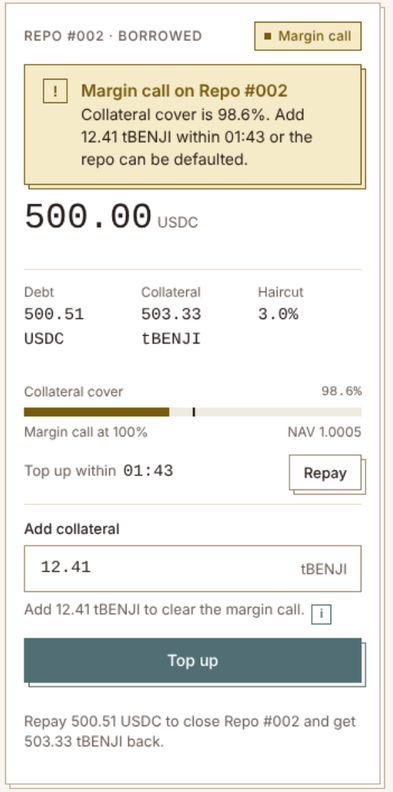
BORROWER RECEIVES
100.00 USDC
from Offer #7

Repo #7 opened. Settled atomically in one transaction

TX 6424…e68d

Settled
REPO 007

2 MARGIN CALL ⟶ TOP UP



3 DEFAULT ⟶ LIQUIDATION

REPO #008 · BORROWED

■ Liquidated

50.00 USDC

Debt	Collateral	Haircut
50.01 USDC	51.55 tBENJI	3.0%

Closed

Liquidated: not repaid in time, so the tBENJI was sold at NAV − 2.0%. The surplus over the debt was paid to this wallet.

TRY IT LIVE

App: caravel-repo.pages.dev/app/

Code: github.com/rcusans-code/Caravel

Caravel app on Stellar testnet, Circle testnet USDC and tBENJI; 1 demo day = 60 s. Screens 1 and 3: 25 Oct 2026, liquidation run by the keeper. Screen 2: the team's seeded testnet run.

One core for every issuer’s compliance

- **Caravel needs only SEP-41 `transfer`.** The issuer decides who may hold the token.
- **BENJI** admits holders with authorized trustlines; **Spiko** with its own Soroban permission manager (Halborn audit: all findings fixed).
- **Risk per asset:** haircut, NAV staleness and on/off, each one admin call.
- **Weekend NAV:** staleness of about 96h per asset. A stale NAV pauses only new repos and margin calls; repay and top up always work.

ELIGIBLE COLLATERAL, 7-DAY TERM

ASSET	HAIRCUT	WHY
USTBL, BENJI, USDY	3%	US T-bill funds; NAV moves by hundredths of a percent a week
EUTBL	~5%	EUR/USD risk against a USDC loan
CETES	8–10%	Mexican sovereign and MXN risk
SAFO	After due diligence only	Returns come from a bank swap: counterparty risk to assess first

Sources: Caravel DECISIONS #24–26; spiko-tech/stellar-contracts (token, permission manager); Halborn audit of Spiko’s Stellar contracts, 16–18 Sep 2025, all fixed by 21 Sep 2025; spiko.io.

The assets already live here

Where the RWAs are	`$3.03B` of tokenized RWAs, incl. Spiko, Ondo and Franklin Templeton.
Authorized trustlines	Compliance at the asset layer. Our spike: an issuer can authorize a contract as a holder.
Stellar Asset Contracts	Classic USDC and tBENJI used straight from Soroban. No wrapped tokens.
Atomic DvP	One signed `take_offer` moves both legs, at the exact collateral amount.
SEP-40 oracles	Reflector’s price interface: swapping the feed is config, not code. Sub-cent base fees.

Testnet build: Circle testnet USDC, tBENJI with AUTH_REQUIRED, a Spiko-style allowlist token (tSAFO) and a mock SEP-40 NAV feed.

Sources: DefiLlama RWA, Stellar chain page, Active AUM, 25 Oct 2026; Caravel spikes/auth (AUTH_REQUIRED spike: GO); Caravel README, “Stellar features used”.

The market already borrows against tokenized funds

26 Aug 2025	**Aave Horizon** launches · a permissioned market to borrow stablecoins against RWAs
14 Nov 2025	**BlackRock BUIDL** on Binance · accepted as off-exchange collateral for trading
11 Feb 2026	**Franklin Templeton** with Binance · Benji tokenized MMFs as off-exchange collateral

SPIKO, THE LARGEST ISSUER ON STELLAR

> “Yes, you can use them for example as collateral to take out on-chain loans. Please contact the Spiko team if you are interested in this use case.”
>
> spiko.io, on using its fund shares in DeFi

Sources: Aave blog, Horizon launch (26 Aug 2025); Securitize via PR Newswire (14 Nov 2025); Franklin Templeton press release (11 Feb 2026); spiko.io/use-cases/web3 (read 25 Oct 2026).

Pools lend open-ended; repo stays bilateral

	TERM	PRICING AND LIQUIDATION	PERMISSIONED FUND SHARES	SETTLEMENT
Blend	Open-ended, variable rate	Oracle price, liquidated any hour	No allowlist hook	On Stellar
Templar	Open-ended	Oracle price, liquidated any hour	Wrapped assets only	Markets on NEAR, bridged
Aave Horizon	Open-ended, variable rate	Oracle price, whitelisted liquidators	Yes, issuer-allowlisted	On Ethereum
XOXNO · Liqvid X on Alula	Open-ended pool	Oracle NAV, liquidated at HF < 1	Spiko spoke built, disabled; Liqvid X in SCF #45 build	On Stellar
CEX collateral	Open-ended	Exchange risk engine	Yes, via a custodian	Off-exchange, off-chain
Caravel	**Fixed term, fixed rate**	**Daily NAV, margin window**	**Yes, the issuer admits the contract**	**Atomic DvP on Stellar**

Ascend × M1X (Oct 2026): a bilateral repo on one sovereign token, rolled hourly. Caravel: an open, multi-issuer order book, daily-NAV margin.

Model: a protocol fee taken from interest only. The lender gets principal + interest − fee, never more than the offer terms.

Sources: Caravel RWA gap analysis: Blend, Templar, Aave Horizon; XOXNO mainnet market config (23 Oct 2026); SCF #45 Liqvid X submission; Binance collateral programs (slide 9); stellar.org press, Ascend × M1X (23 Oct 2026); DECISIONS #3.

First deals OTC, then open the desk

Go-to-market

First deals arranged OTC with one issuer, **Spiko or Etherfuse**, and one market maker as the approved liquidator.

Launch collateral: T-bill funds at a 3% haircut (slide 7). Spiko says its shares can be used as on-chain collateral and asks partners to get in touch (slide 9).

Roadmap

1. Real NAV oracle
2. Liquidation auction
3. Lender vault

Also on the list: NAV bounds per asset, split admin keys.

Team of four

Denys Nichoha
Contracts: core flows

Misha
Integration: oracle, keeper, deploy

Bohdan Chernenko
Design and frontend

Bohdan Koldun
Pitch, docs, QA

Code `github.com/rcusans-code/Caravel`
App `caravel-repo.pages.dev/app/`

Risks we know and how we fix them:

- **Lender cold start:** tradable repo-claim token (Term-style) + park untaken offers in Blend pool.
- **Unproven fixed-term borrower:** treasuries first, MMs second; liquidation on fund's clock; atomic roll.
- **Bad debt in margin window:** 2.5% early margin trigger, 50–100 bps liquidation discount, NAV bounds.

Sources: Caravel pitch plan (docs/pitch-deck.md); README, “Risks and limits” and roadmap.